AI and renewable energy across U.S. states, explained

This notebook explains a study of AI (artificial intelligence) and renewable energy across U.S. states.

ECON 499R, Emory University, Fall 2026. Shunji Lewandowski and Alexander Hendricks; advisor Prof. Fohlin. Written for a reader who hasn't taken econometrics. Every number in this document is read from the project's result files and logged in `results/2026-11-09/explained/number_check.csv`.

Everything here is an association. Our data and methods can show that two things move together. They can't show that one causes the other, so we never say AI "causes," "drives" or "raises" anything.

0. One-page summary

The question. Do U.S. states where AI-related activity grows faster also see more or less electricity use, energy efficiency and renewable energy? We test two recent papers on the same data: Zhang et al. (2026), "Blessing or Peril?", which studied 261 Chinese cities, and Fang, Lee and Li (2025), which found that U.S. states with more AI job postings used more renewable energy.

Our AI measure is private data-center jobs per 1,000 workers, from BLS (Bureau of Labor Statistics) records of jobs in NAICS 518210 (North American Industry Classification System code 518210, Computing Infrastructure Providers, Data Processing, Web Hosting). We use all 50 states from 2015 to 2025, which gives 550 state-years in our main models.

The five main findings.

1. **Electricity use and efficiency: no association.** A state's data-center jobs aren't associated with its electricity use per person (−0.00157, p = 0.795) or with how much economic output it gets per unit of electricity (0.00221, p = 0.707).
2. **Renewable share: a negative association that fails a timing test.** More data-center jobs go with a slightly lower renewable share (−1.01 percentage points per extra job per 1,000 workers, p = 0.100). But *next* year's AI jobs are associated with *this* year's share even more strongly (−1.97 points, p = 0.006). A cause can't come after its effect, so this points to where and when data-center jobs grow, rather than AI lowering the renewable share.
3. **New renewable capacity: positive in raw megawatts, but not robust.** States add 70.8 MW (megawatts) more renewable capacity per extra job per 1,000 workers (p = 0.033), but the result disappears once we account for the size of each state's grid (p = 0.616), and future AI jobs "predict" it too.
4. **No channel and no spillover.** AI jobs aren't associated with the share of tech jobs, the channel Zhang et al. propose (p = 0.454). A spillover onto neighboring states' renewable share (−16.1 points per extra job per 1,000 workers, about −1.4 points for a typical within-state swing; p = 0.035) also fails the timing test.
5. **Fang's positive result depends on the data version and on leaving out year effects.** On today's EIA (U.S. Energy Information Administration) data, Fang's main coefficient (0.0531) becomes −0.0188 (p = 0.317). On the data as published before EIA changed how it counts wind and solar, it is 0.0348 (p = 0.057). Once we allow for nationwide trends, it is never significantly positive; the estimates are near zero or negative.

What they mean. Over 2015-2025, we find no sign that data-center growth went with lower efficiency, higher electricity use or a lower renewable share once timing is taken into account. Neither the "blessing" nor the "peril" holds up in state data.

What we can say: a typical within-state swing in data-center jobs (one SD (standard deviation), 0.521 jobs per 1,000 workers) isn't associated with a fall in electricity use of more than 0.71% or a rise of more than 0.54%, or with a fall in the renewable share of more than 1.2 percentage points.

What we can't say: that AI has no effect. Jobs are a rough stand-in for data-center power use, most of the data-center boom comes after our data end, and state totals can't see national or cross-state effects. Georgia Power alone reports large-load commitments of 15,600 MW across 32 customers as of June 30, 2026, for load scheduled through the mid-2030s, 8,700 MW of it by winter 2028/2029 (Q2 2026 large-load report to the Georgia PSC (Public Service Commission), p. 1 [PDF p. 5]); our data end in 2025.

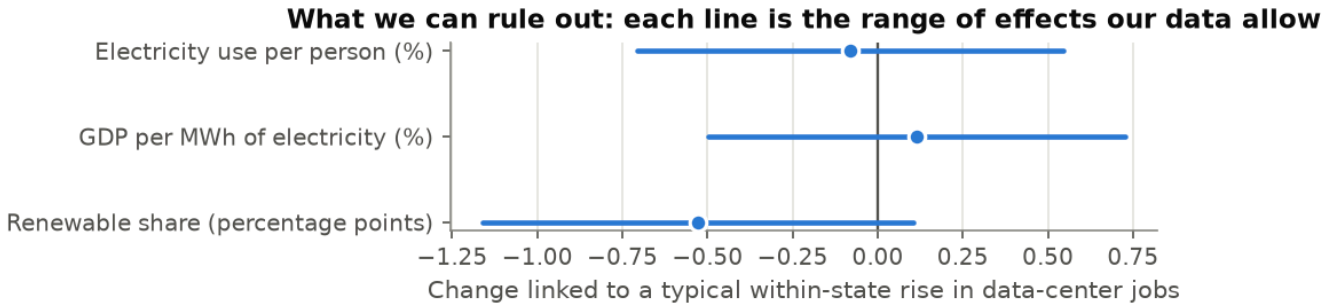


Figure 1. Estimate (dot) and 95% confidence interval (line) per one within-state standard deviation of data-center jobs. Every interval crosses zero, and each is narrow enough to rule out large effects.

1. How to read this notebook

Three ideas carry almost every result: a coefficient, a confidence interval and a p-value. We use one real result, the renewable share, as the running example.

A coefficient is a slope. It says how much the outcome tends to change when AI changes by one unit, holding the other things in the model fixed. Our AI unit is one extra data-center job per 1,000 workers. For the renewable share, the coefficient is −0.0101: a state with one more data-center job per 1,000 workers tends to have a renewable share 1.01 percentage points lower. *Analogy: it's the "price per pound" on a deli scale. It tells you how much one more unit adds, not the total bill.*

A confidence interval is the range of slopes the data can't rule out. We report 95% intervals. For the renewable share it runs from −2.2 to 0.20 percentage points. *Analogy: a weather forecast that says "between 60 and 70 degrees." The narrower the range, the more useful the forecast. If the range includes zero, "no effect" is one of the possibilities the data allow.*

A p-value measures surprise. It's the chance of seeing a slope at least this far from zero if the true slope were zero. Small values (below 0.05 by convention) mean "this would be surprising if nothing were going on." For the renewable share, p = 0.100: not small enough by the usual standard. *Analogy: a friend flips a coin and gets heads 6 times out of 10. That's not surprising for a fair coin. Heads 10 times out of 10 would be. A p-value says which of those situations you're in.*

What a p-value isn't. It isn't the chance that the result is true, and a large p-value doesn't prove there's no effect. That's why we also report what the confidence interval rules out.

How to read the charts. Coefficient charts show a dot for the estimate and a line for its 95% confidence interval, with a gray vertical line at zero. If the line crosses zero, the result isn't statistically different from zero.

2. Key ideas, each with an analogy

Table 1. Fourteen ideas used throughout.

Idea	Analogy and why it matters
Correlation is not causation	Ice-cream sales and sunburns both rise in summer. Neither causes the other: the sun drives both. Two state trends can move together for the same reason. That's why everything here is an *association*.
Panel data	We follow the same 50 states every year from 2015 to 2025, like 550 report cards: one per student per year. Seeing each state many times lets us compare a state to itself.
State fixed effects	Compare each student to their own past grades, not Alaska to Texas. Texas always uses more electricity than Vermont for reasons unrelated to AI (size, climate, industry). State fixed effects remove each state's permanent level, so only changes within a state count.
Year fixed effects	Grade on a curve each year. Some things hit every state at once: federal tax credits for wind and solar, COVID in 2020, falling solar prices. Year fixed effects remove each year's nationwide average, so a national trend can't masquerade as an AI effect.
Lagging AI one year	The cause should come before the effect. We use last year's AI jobs to explain this year's outcome, so the order in time is at least right.
Lead or placebo test	A forecast that only works after it rains isn't a forecast. If *next* year's AI jobs "predict" *this* year's renewables as well as last year's do, the association can't be AI acting on renewables. Something else, like where data centers choose to locate, is going on.
Clustered standard errors	Asking the same person a question 11 times isn't 11 independent opinions. A state's 11 years are related to each other, so we treat them as one cluster. This makes our confidence intervals honestly wider.
Logs and percentage changes	Many outcomes are in logs. A change of 0.01 in a log is about a 1% change, so log results read as percentages whatever the state's size.
A precise null	A bathroom scale that's accurate to one pound and reads zero tells you something: there's nothing heavy on it. A broken scale that reads zero tells you nothing. A null result with a narrow confidence interval is the first kind: it rules out large effects.
Multiple testing	Flip enough coins and some will land heads five times in a row. Run 30 tests and one or two will look significant by luck. We use the Benjamini-Hochberg correction, which raises the bar when many tests are run together, and report its q-value.
Placebo industries	If pet-care jobs "predict" renewables as well as AI jobs do, the model is picking up shared growth trends, not anything specific to AI. Pet care has nothing to do with electricity.
Spatial spillovers	Drop a stone in a pond and the ripples reach the shore. AI growth in one state might ripple into its neighbors through shared power grids and markets. Spatial models measure those ripples.
The 2023 EIA (U.S. Energy Information Administration) accounting change	Imagine changing the exchange rate used to convert foreign coins into dollars. In 2023, EIA changed how SEDS (State Energy Data System) converts wind and solar electricity into energy units, which made renewable energy look smaller in every year. Same wind, different yardstick.
Learning curves	TVs and solar panels get cheaper as more are made, because factories learn and scale up. A learning rate of 20% means each doubling of total production cuts the cost by 20%.

3. The data

We built a state-by-year panel from public sources. The table lists the main variables; the full list, with every source file and its checks, is in `data/variable_registry.csv`.

Table 2. Main variables. Years and units are read from the variable registry.

Variable	What it measures	Source	Years	Units	Why we need it
AI measure: data-center jobs	Private jobs in data processing and hosting per 1,000 workers, last year's value	BLS (Bureau of Labor Statistics) QCEW (Quarterly Census of Employment and Wages), NAICS 518210 (North American Industry Classification System code 518210, Computing Infrastructure Providers, Data Processing, Web Hosting)	2015-2025	per 1,000 workers	Our main AI measure
AI job postings	Online job ads asking for AI skills, by state	Stanford AI Index (from vendor job-ad data)	2018-2025	count	Fang's AI measure
Electricity use	Retail electricity sales per person, in logs	EIA (U.S. Energy Information Administration) sales data; BEA (Bureau of Economic Analysis) population	2015-2025	log	Outcome: consumption
Energy efficiency	Real GDP (gross domestic product) per MWh (megawatt-hours) of electricity, in logs	BEA GDP; EIA sales	2015-2025	log	Outcome: efficiency
Renewable share	Share of in-state utility-scale generation from renewables	EIA generation by state	2015-2025	share	Outcome: clean power
New renewable capacity	Change in renewable capacity from the year before	EIA capacity by state	2015-2025	MW	Outcome: build-out
Renewable energy use	Renewable energy consumed in the state	EIA SEDS (State Energy Data System)	2015-2024	billion Btu (British thermal units)	Fang's outcome
Income	Real GDP per person, in logs	BEA	2015-2025	log	Control
Manufacturing	Manufacturing share of GDP	BEA	2015-2025	share	Control

Variable	What it measures	Source	Years	Units	Why we need it
Electricity price	Average retail price	EIA	2015-2025	cents/kWh (kilowatt-hour)	Control
Population growth	Growth from the year before	BEA	2015-2025	share	Control
Tech jobs	Computer-systems-design jobs as a share of all jobs	BLS QCEW, NAICS 5415	2015-2025	share	Zhang's channel (mediator)
Economic density	Nonfarm GDP per square mile, in logs	BEA; Census Bureau land area	2015-2025	log	Second channel (mediator)
Policy uncertainty	News-based index of state economic policy uncertainty	EPU (Economic Policy Uncertainty index) by Baker, Davis and Levy	1985-2025	index	Fang's moderator
Renewable mandates	Each state's RPS (Renewable Portfolio Standard) or CES (Clean Energy Standard) target for the year	LBNL (Lawrence Berkeley National Laboratory) RPS and CES data	2015-2025	percent	Control in a check
Weather	Heating and cooling degree days	NOAA (National Oceanic and Atmospheric Administration) HDD (heating degree days) and CDD (cooling degree days)	2015-2025	degree days	Control in a check
Federal tax credits	Investment Tax Credit rate for solar (ITC (Investment Tax Credit)); wind PTC (Production Tax Credit) similar	CRS (Congressional Research Service) reports and federal law	2015-2025	percent	National driver
Solar cost	LCOE (levelized cost of energy) of new utility-scale solar	LBNL Utility-Scale Solar	2015-2025	2025 $/MWh	National driver; Georgia
AI research workers	Computer and information research scientists	BLS OEWS (Occupational Employment and Wage Statistics)	2015-2024	jobs	AI index

Also used: BTOS (Business Trends and Outlook Survey), the Census Bureau's survey of the share of firms using AI, as a check on our AI index; NCSES (National Center for Science and Engineering Statistics) degree data, NSF (National Science Foundation) R&D (research and development), ACS (American Community Survey) broadband and Epoch AI data-center data for side analyses; and Georgia PSC (Public Service Commission) filings. USPTO (U.S. Patent and Trademark Office) patent data were downloaded and not used.

Why so many sources? No single agency measures AI, electricity and policy together, so each source answers one piece of the question: BLS (Bureau of Labor Statistics) counts the jobs, EIA (U.S. Energy Information Administration) measures the electricity, BEA (Bureau of Economic Analysis) measures the economy, and LBNL (Lawrence Berkeley National Laboratory), CRS (Congressional Research Service) and NOAA (National Oceanic and Atmospheric Administration) supply the mandates, tax credits and weather that could otherwise be mistaken for AI. Each is the official or standard source for its piece.

4. The Zhang test: our main design

Zhang et al. (2026) propose three hypotheses, which we test on U.S. states: **H1**, AI is associated with electricity use, efficiency and renewables; **H2**, the association runs through channels such as tech jobs or economic density; **H3**, AI in one place spills over to its neighbors. We use their design where U.S. data allow.

4.1 H1: two-way fixed effects

The question. When a state's data-center jobs rise faster than usual, do its electricity use, efficiency, renewable share or renewable build-out also change, compared with its own past and with the national trend?

$$y_{st} = \beta\, AI_{s,t-1} + \gamma' X_{st} + \alpha_s + \delta_t + \varepsilon_{st}$$

Table 3. Symbols in the H1 equation.

Symbol	Plain meaning
y_{st}	The outcome in state s, year t: electricity use, efficiency, renewable share or new renewable capacity
$AI_{s,t-1}$	Last year's private data-center jobs per 1,000 workers (NAICS 518210 (North American Industry Classification System code 518210, Computing Infrastructure Providers, Data Processing, Web Hosting))
β	The number we care about: the association between AI and the outcome
X_{st}, γ	The four controls (log real GDP per person, manufacturing share of GDP, electricity price, population growth) and their associations with the outcome
α_s	State fixed effect: each state's permanent level
δ_t	Year fixed effect: what hit every state in year t
ε_{st}	Everything else, which we treat as noise; standard errors are clustered by state

Analogy. A teacher wants to know whether extra tutoring helps grades. She compares each student with their own past grades (state effects), curves each year's grades (year effects), and asks whether students who got more tutoring *last* term improved more *this* term.

What "yes" and "no" would look like. "Yes" (Zhang's prediction): a coefficient clearly away from zero, with a confidence interval that excludes zero, and a lead test that comes back empty. "No": a coefficient near zero with a narrow interval that excludes large values.

Table 4. H1 results: last year's data-center jobs per 1,000 workers, state and year fixed effects, four controls, standard errors clustered by state, 50 states, 2015-2025.

Outcome	Coefficient	95% confidence interval	p-value	State-years
Electricity use per person (log)	−0.00157	−0.0136 to 0.0105	p = 0.795	550
GDP per MWh (log)	0.00221	−0.00953 to 0.0139	p = 0.707	550
Renewable share (0-1)	−0.0101	−0.0223 to 0.00202	p = 0.100	550
New renewable capacity, MW	70.8	5.85 to 136	p = 0.033	550
Same, per 1,000 MW of 2015 capacity	0.996	−2.98 to 4.97	p = 0.616	550

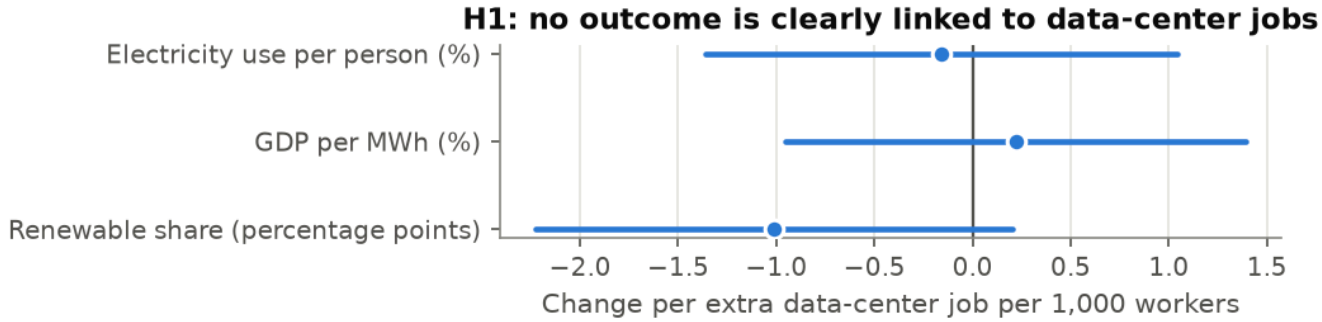


Figure 2. H1 coefficients times 100, with 95% confidence intervals. Log outcomes read as percent; the renewable share reads as percentage points.

In plain English. States whose data-center jobs grew faster didn't use more or less electricity per person, and didn't get more or less economic output per unit of electricity, than their own trends predict. The renewable share is slightly lower (p = 0.100), and new renewable capacity is higher in raw MW (p = 0.033), but section 4.4 shows that neither survives a timing test.

What we can rule out. Per typical within-state rise in data-center jobs (one SD), the 95% interval for electricity use runs from −0.71 to 0.54%, for output per MWh from −0.50 to 0.73%, and for the renewable share from −1.2 to 0.11 percentage points. So we can rule out, for example, a 1% rise in electricity use per person from a typical swing in data-center jobs.

4.2 H2: mediation (is there a channel?)

The question. If AI were associated with the outcomes, would it be *through* something else, such as more tech jobs (Zhang's channel), a combined AI-and-tech index, or denser economic activity?

$$M_{st} = a\, AI_{s,t-1} + \text{controls} + \alpha_s + \delta_t + u_{st}$$

$$y_{st} = c'\, AI_{s,t-1} + b\, M_{st} + \text{controls} + \alpha_s + \delta_t + \varepsilon_{st}$$

$$\text{indirect association} = a \times b$$

Table 5. Symbols in the mediation equations.

Symbol	Plain meaning
M_{st}	The proposed channel (mediator): tech-job share, the AI-tech coupling index, or economic density
a	Is AI associated with the channel? (the first link)
b	Is the channel associated with the outcome, holding AI fixed? (the second link)
c'	What's left of the AI association after the channel is accounted for
$a \times b$	The part of the association that runs through the channel

Analogy. A bucket brigade passes water from a well to a fire. Water reaches the fire through the brigade only if both handoffs work: well to person (a) and person to fire (b). If either link is broken, nothing flows through that path.

What "yes" and "no" would look like. "Yes": both links are clearly nonzero and the product $a \times b$ has a confidence interval that excludes zero. "No": the first link fails, so nothing can flow through it.

Table 6. H2 first links, with the Sobel test range across all 9 channel-outcome pairs below.

Channel	First link: AI to channel (a)	p-value	Comment
Tech-job share (Zhang's channel)	−0.000157	p = 0.454	Fails: AI isn't associated with the tech-job share
AI-tech coupling index	0.0416	p < 0.001	Positive by construction: the index is built from AI itself
Economic density	−0.00336	p = 0.296	Fails: income already absorbs it

Sobel p-values across the 9 pairs run from 0.322 to 0.991. Bootstrap: 999 cluster resamples.

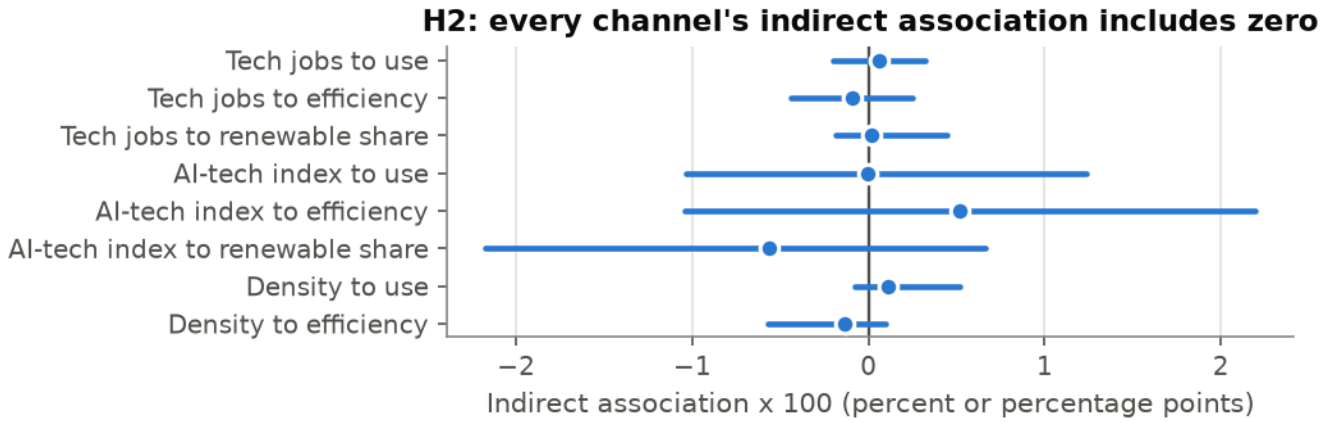


Figure 3. Indirect association (a times b) for each channel and outcome, with 95% cluster-bootstrap intervals.

In plain English. Data-center jobs aren't associated with tech jobs in general (p = 0.454), so Zhang's channel has nothing to carry. The coupling index looks linked only because it's built from the AI measure itself. Every indirect association has a confidence interval that includes zero.

What we can rule out. For the tech-job channel, the first link's interval runs from −0.00057 to 0.00026: a typical swing in data-center jobs moves the tech-job share by far less than a hundredth of a percentage point.

4.3 H3: spatial spillovers

The question. Do states cluster by renewable share, and is AI growth in *neighboring* states associated with a state's own outcomes?

Moran's I (do neighbors look alike?):

$$I = \frac{N}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij}(y_i - \bar{y})(y_j - \bar{y})}{\sum_i (y_i - \bar{y})^2}$$

SLX (spatial lag of X model) model (neighbors' AI on the right-hand side), and **SDM (spatial Durbin model)** (also neighbors' outcomes):

$$y_{st} = \rho \sum_j w_{sj}\, y_{jt} + \beta\, AI_{s,t-1} + \theta \sum_j w_{sj}\, AI_{j,t-1} + \gamma' X_{st} + \alpha_s + \delta_t + \varepsilon_{st}$$

The SLX model is the same equation with $\rho = 0$.

Table 7. Symbols in the spatial models.

Symbol	Plain meaning
w_{ij}	How close states i and j are. Main version: one over the distance between state centers, all 50 states; check: shared border, 48 contiguous states. Each state's weights sum to one
$\sum_j w_{sj} AI_{j,t-1}$	The average AI of state s's neighbors, closer ones counting more
θ	Association with neighbors' AI
ρ	How much a state's outcome moves with its neighbors' outcomes (the echo)
Direct effect	Association of a state's own AI with its own outcome, including echoes back from neighbors
Indirect effect	Association of all other states' AI with this state's outcome: the spillover
Total effect	Direct plus indirect

Analogy. Drop a stone in a pond (AI growth in one state). The direct effect is the splash where it lands; the indirect effect is the ripples reaching other shores; ρ is how much the ripples bounce back and forth.

What "yes" and "no" would look like. "Yes": a clearly nonzero indirect effect that comes from neighbors' *past* AI, not their *future* AI. "No": no indirect effect, or one that future AI explains as well.

Table 8. H3 results for the renewable share (0-1 scale). SDM (spatial Durbin model) is the main specification (maximum likelihood, inverse-distance weights).

Test (renewable share)	Estimate	95% interval or note	p-value
Moran's I, lowest year	0.0941 (2015)	States with similar shares cluster	p = 0.001
Moran's I, highest year	0.157 (2025)	Clustering rises over time	p = 0.001
SLX (spatial lag of X model): own AI	−0.0136	Clustered by state	p = 0.014
SLX: neighbors' AI	−0.0551	Clustered by state	p = 0.117
SDM: echo	0.617	Main spec, inverse distance	
SDM: direct	−0.0155	−0.024 to −0.0073	p < 0.001
SDM: indirect (spillover)	−0.161	−0.34 to −0.035	p = 0.035
SDM: total	−0.177	−0.36 to −0.046	p = 0.026
Timing, border neighbors: past AI	0.00977	2015-2023, lag and lead together	p = 0.583
Timing, border neighbors: future AI	−0.0406	2015-2023, lag and lead together	p = 0.027

Test (renewable share)	Estimate	95% interval or note	p-value
Timing, inverse distance: past AI	−0.0105	2015-2023, lag and lead together	p = 0.760
Timing, inverse distance: future AI	−0.00633	2015-2023, lag and lead together	p = 0.868

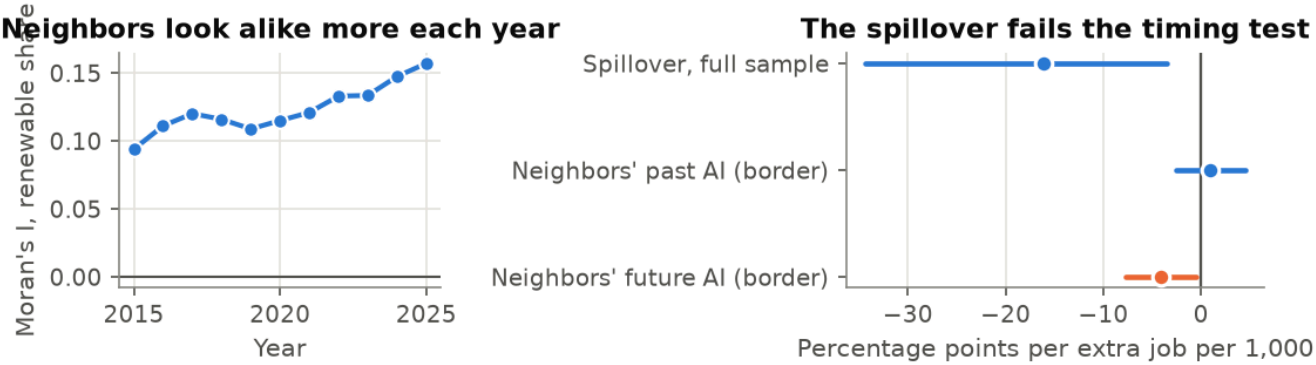


Figure 4. Left: Moran's I for the renewable share each year. Right: the full-sample spillover (spatial Durbin model), and on 2015-2023 the association with neighbors' past and future AI (border weights), with 95% intervals.

In plain English. Neighboring states do have similar renewable shares, and the similarity grows over time. The full-sample model finds a negative spillover (p = 0.035). But when we add neighbors' *future* AI, it's the future AI that carries the association with border weights (p = 0.027), not the past (p = 0.583). A ripple that arrives before the stone is dropped isn't a ripple, so H3 is not supported.

What we can rule out. Per typical within-state swing in neighbors' AI, the full-sample spillover is −1.39 percentage points, with a 95% interval of −3.0 to −0.30 points. That rules out a positive spillover and a negative one much larger than the lower end. Because it can't be separated from reverse timing, we don't read it as a spillover at all.

4.4 Robustness: lead, lags, long difference, state trends

The question. Do the H1 results hold when we change how time enters the model?

$$y_{st} = \beta_{lag}\, AI_{s,t-1} + \beta_{lead}\, AI_{s,t+1} + \text{controls} + \alpha_s + \delta_t + \varepsilon_{st}$$

Table 9. Symbols and variants in the robustness checks. All other symbols are as in H1.

Symbol or variant	Plain meaning
β_{lead}, $AI_{s,t+1}$	Next year's AI: the lead or placebo. It shouldn't matter if AI acts on outcomes
β_{lag}	Last year's AI, estimated alongside the lead
Distributed lags	AI one, two and three years back together (2017-2025), and their sum
Long difference	Change in AI from 2014 to 2024 against the change in the outcome from 2015 to 2025, one row per state
State trends	Each state gets its own straight-line trend, so steady state-specific growth can't masquerade as AI

Analogy. A rooster crows before sunrise, but nobody thinks the rooster raises the sun. Checking whether *next* year's AI "explains" this year's outcome is checking for roosters.

What "yes" and "no" would look like. A real association: last year's AI matters and next year's doesn't. A timing problem: next year's AI matters as much or more.

Table 10. Robustness of the renewable-share association, per extra data-center job per 1,000 workers.

Version (renewable share)	Coefficient (percentage points)	95% interval	p-value
Main H1 (last year's AI)	−1.01	−2.2 to 0.20	p = 0.100
Next year's AI alone (lead)	−1.97	−3.4 to −0.59	p = 0.006
Both together: last year's AI	0.146	−1.2 to 1.5	p = 0.827
Both together: next year's AI	−2.10	−3.6 to −0.59	p = 0.007
Distributed lags, sum of 3 years	−0.950	−2.2 to 0.27	p = 0.124
Long difference, 50 states	−1.37	−3.1 to 0.32	p = 0.110
Drop 2020	−1.08	−2.3 to 0.16	p = 0.087
Drop Virginia	−1.01	−2.2 to 0.22	p = 0.104
Control for RPS (Renewable Portfolio Standard) targets	−1.03	−2.3 to 0.20	p = 0.099
All-ownership AI jobs	−0.996	−2.2 to 0.20	p = 0.102
With state trends	0.596		p = 0.196

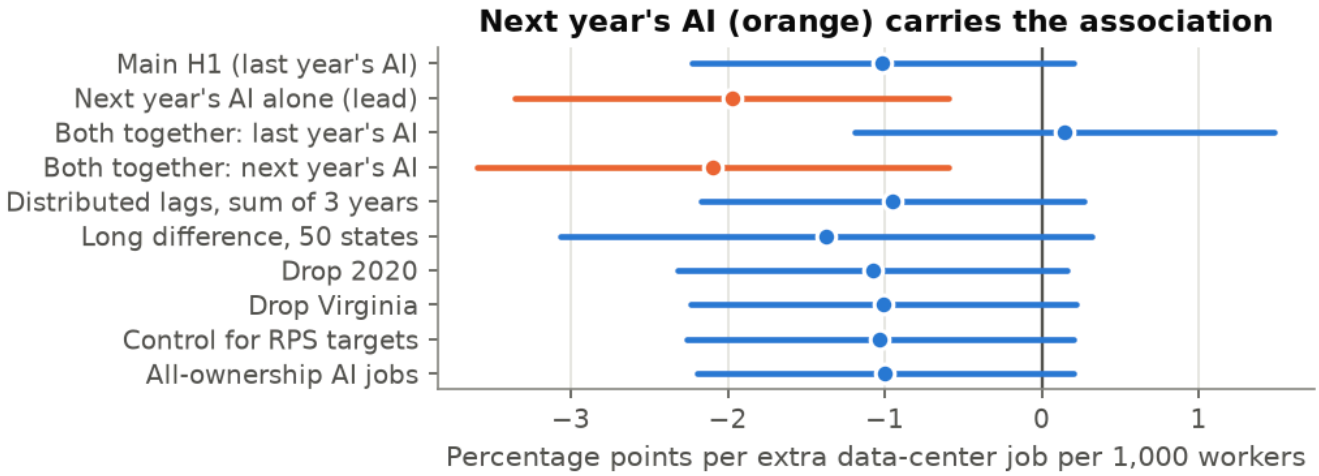


Figure 5. Renewable-share coefficient under each version, with 95% intervals. Orange rows use next year's AI.

In plain English. Every version gives about the same small negative number, but the lead test breaks the story: next year's AI is associated with this year's renewable share (p = 0.006), and with both in the model, last year's AI drops to about zero (0.146 points, p = 0.827). The negative association points to where and when data-center jobs grow, rather than AI lowering the share. New renewable capacity has the same problem: its lead is also positive (86.8 MW, p = 0.028).

Two more checks change nothing. Assigning the Gavins Point dam (on the Nebraska-South Dakota border) to a single state gives 71.0 MW for new capacity. Adding NOAA (National Oceanic and Atmospheric Administration) HDD (heating degree days) and CDD (cooling degree days) to the electricity-use model on the 48 contiguous states gives −0.00211 (p = 0.711), against −0.000888 without them.

What we can rule out. With both lead and lag in the model, the interval for last year's AI is −1.2 to 1.5 percentage points per extra job per 1,000 workers.

4.5 Instrumental variable: why it was uninformative

The question. Can we isolate AI growth that comes from national industry growth rather than from local conditions, to get closer to cause and effect?

$$\text{First stage: } AI_{s,t-1} = \pi\, Z_{s,t-1} + \text{controls} + \alpha_s + \delta_t + u_{st}$$

$$Z_{s,t} = \text{share}_{s,2015} \times \text{national growth}_t$$

$$\text{Second stage: } y_{st} = \beta_{IV}\, \widehat{AI}_{s,t-1} + \text{controls} + \alpha_s + \delta_t + \varepsilon_{st}$$

Table 11. Symbols in the IV (instrumental variable) model.

Symbol	Plain meaning
$Z_{s,t}$	The instrument: the state's 2015 share of national data-center jobs times national data-center growth (a shift-share)
π	How strongly the instrument predicts actual AI jobs
$\widehat{AI}$	The part of AI that the instrument predicts
β_{IV}	The association using only that predicted part
First-stage F	Instrument strength; the usual rule of thumb asks for at least 10

Analogy. To learn whether fertilizer helps crops, you'd like fertilizer handed out by coin flip. The instrument is a stand-in coin flip. If the "coin" barely changes who gets fertilizer, it can't tell you anything.

What "yes" and "no" would look like. A useful instrument has a first-stage F above 10 and a reasonably tight second-stage interval. A weak one has F below 10 and a second stage too wide to read.

Table 12. Shift-share IV (instrumental variable), 2015 base shares, clustered by state.

Quantity	Estimate	95% interval	p-value
First-stage F (instrument strength)	2.81	Needs 10 or more	
Renewable share, ordinary regression	−0.0101	−0.022 to 0.0020	p = 0.100
Renewable share, IV	0.112	−0.075 to 0.30	p = 0.236
Electricity use, IV	0.0392	−0.11 to 0.19	p = 0.596

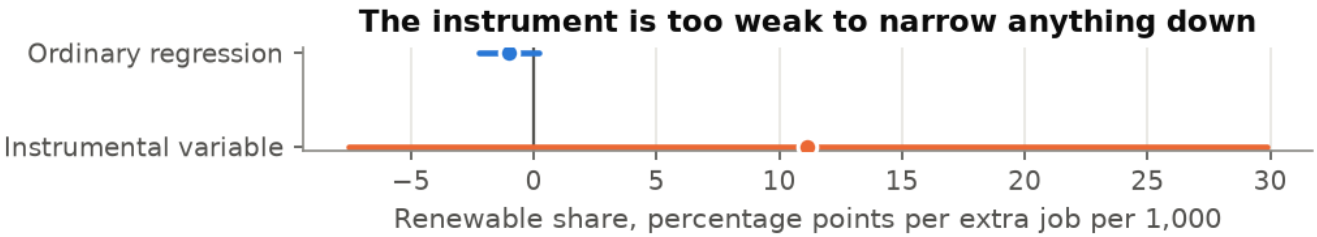


Figure 6. Renewable share: ordinary regression vs instrumental variable, with 95% intervals.

In plain English. The instrument barely predicts actual data-center jobs (F = 2.81, well below 10), so the IV (instrumental variable) estimate's interval is many times wider than the ordinary one. It can't confirm or reject anything, and we treat it as uninformative.

What we can rule out. Nothing beyond what the ordinary regression already rules out.

4.6 Nonlinearity: is there a turning point?

The question. Is the association curved, for example rising at low AI levels and falling at high ones, as Melguizo et al. (2026) find across countries?

$$y_{st} = \beta_1 \, AI_{s,t-1} + \beta_2 \, AI^2_{s,t-1} + \text{controls} + \alpha_s + \delta_t + \varepsilon_{st}$$

$$\text{slope at AI level } x: \quad \beta_1 + 2\beta_2 x$$

Table 13. Symbols in the nonlinearity model.

Symbol	Plain meaning
β_2	Curvature: negative means an upside-down U, positive means a U
$\beta_1 + 2\beta_2 x$	The slope at AI level x
Lind-Mehlum test	A real curve needs the slope to be clearly positive at one end of the data and clearly negative at the other (one-sided t above 1.645 in size at both ends)

Analogy. A hill has to go up on one side and down on the other. A flat field with a slight tilt isn't a hill, even if a curved line fits it.

What "yes" and "no" would look like. "Yes": significant slopes of opposite sign at the lowest and highest AI levels. "No": slopes near zero at one or both ends.

Table 14. AI squared, H1 design. Read from the corrected file results/2026-11-09/h1/lm_proper.csv.

Outcome	Curvature β_2	p-value	t at lowest AI	t at highest AI	Passes?
Electricity use	−0.00299	p = 0.364	0.84	−0.91	No
Output per MWh	0.000946	p = 0.817	−0.14	0.33	No
Renewable share	0.00107	p = 0.818	−0.62	−0.20	No
Solar share	−0.000501	p = 0.710	0.39	−0.31	No
Wind share	−0.000713	p = 0.864	−0.17	−0.54	No
New renewable MW	−8.53	p = 0.606	1.2	0.33	No
Fossil to renewable ratio	−0.334	p = 0.234	1.7	−0.43	No
Coal share	−0.00138	p = 0.771	0.35	−0.20	No

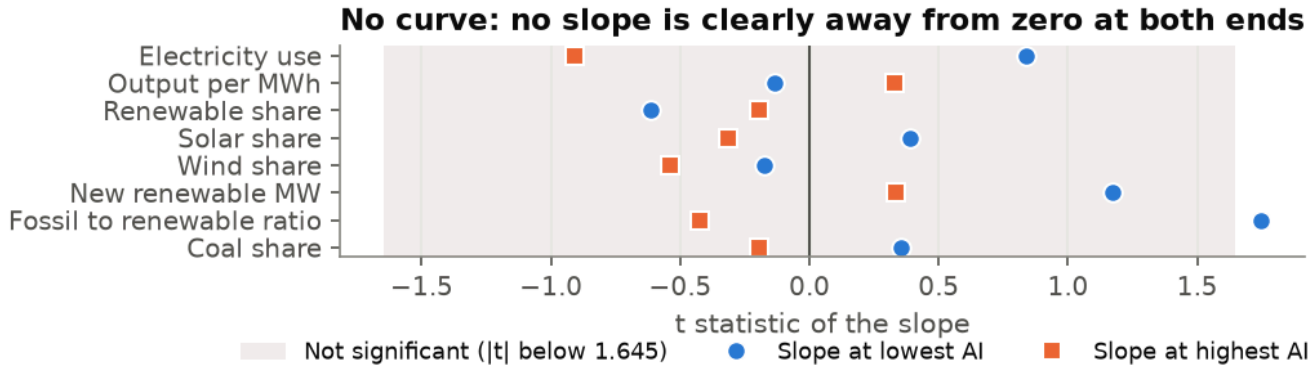


Figure 7. t statistics of the slope at the lowest and highest AI levels. A curve needs both points outside the gray band, on opposite sides.

In plain English. None of the eight outcomes shows a curve. The slopes at both ends are within the "could be zero" band for every outcome, so there's no turning point to report.

What we can rule out. Large curvature: for electricity use, β_2's interval is −0.0096 to 0.0036.

5. The Fang test

Fang, Lee and Li (2025) report that U.S. states with more AI job postings used more renewable energy from 2018 to 2021. We ask three things: can we reproduce their number, why does it differ on today's data, and does it survive our design? Their AI measure is postings from the Stanford AI Index; their outcome is renewable energy use from EIA (U.S. Energy Information Administration) SEDS (State Energy Data System).

5.1 Replication on each EIA data version

The question. Do we get Fang's coefficient with Fang's own model?

$$\ln RE_{st} = \beta \ln AI_{st} + \lambda \ln EPU_{st} + \gamma' X_{st} + \alpha_s + \varepsilon_{st}$$

Table 15. Symbols in Fang's model (their Table 2, column 6).

Symbol	Plain meaning
$\ln RE_{st}$	Renewable energy use from EIA (U.S. Energy Information Administration) SEDS (State Energy Data System), in logs
$\ln AI_{st}$	This year's AI job postings, in logs (no lag)
$\ln EPU_{st}$	State EPU (Economic Policy Uncertainty index), in logs
X_{st}	Fang's controls: energy price, GDP, population, education
α_s	State fixed effects. **No year fixed effects**, unlike our design

Analogy. Re-running a recipe from a cookbook with the same ingredients to check you get the same cake, then trying again with today's flour, which the mill now measures differently.

What "yes" and "no" would look like. A successful replication lands inside Fang's confidence interval; a failed one doesn't.

Table 16. Fang's Table 2, column 6, on each SEDS (State Energy Data System) version, 2018-2021 with DC (District of Columbia), 204 state-years.

EIA (U.S. Energy Information Administration) data version	AI coefficient	95% interval	p-value
Fang's published value	0.0531		
Today's EIA SEDS	−0.0188	−0.056 to 0.018	p = 0.317
SEDS as published in June 2023 (before the change)	0.0348	−0.0011 to 0.071	p = 0.057
Our year-by-year rebuild of the old method	0.0137	−0.026 to 0.054	p = 0.500
Old method, rough scaling (k = 2.6, check only)	0.0216	−0.018 to 0.062	p = 0.287

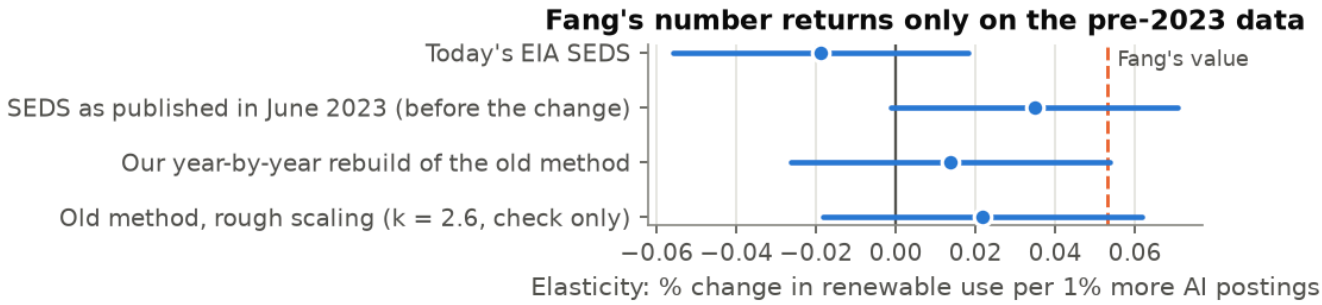


Figure 8. Fang's column 6 coefficient on each EIA data version, with 95% intervals; the dashed line is Fang's published value.

In plain English. On today's data, the coefficient is negative and not significant. On the data EIA published in June 2023, before its accounting change, we get about two-thirds of Fang's number (0.0348, p = 0.057), and columns 1 and 2 come close to Fang's (0.0346 and 0.0435 against 0.0428 and 0.0452). Of the gap between today's and the June 2023 data in column 6, 61% comes from the accounting change and the rest from EIA's later data revisions.

What we can rule out. On today's data, an elasticity above 0.018: a 10% rise in postings isn't associated with more than about 0.18% more renewable use.

5.2 Table A1 check: our data match Fang's

The question. Before blaming the model, are we using the same data? Fang's supplement (Table A1) lists the mean, spread, minimum and maximum of each variable, so we compare ours line by line. The model is a plain comparison, so instead of an equation the check is: difference in means, measured in standard deviations, with differences above 0.25 SD flagged.

Table 17. Our 2018-2021 data against Fang's Table A1, 204 state-years each.

Variable	Our mean	Fang's mean	Difference in our SDs	Reading
ln AI postings	7.701	7.700	0.00056	Matches
ln energy price	2.927	2.930	−0.013	Matches
ln population	8.272	8.270	0.0024	Matches
ln education	4.232	4.230	0.0055	Matches
ln GDP	12.35	12.26	0.086	GDP base year: 2017 vs 2012 dollars
ln EPU (Economic Policy Uncertainty index)	5.382	6.090	−1.2	Lower on average, but not by a constant ratio (see text)
ln renewables, today's SEDS (State Energy Data System)	11.44	11.79	−0.32	Today's data sit below Fang's
ln renewables, June 2023 SEDS	11.83	11.79	0.038	Matches: Fang used the old data

Pending: a rerun of Fang's model with GDP in chained 2012 dollars (BEA (Bureau of Economic Analysis)'s earlier base year). BEA's archive of earlier releases lists its files only in a web browser, so the data couldn't be fetched by script; a person needs to download them. The 0.09 log-point GDP gap is the size a base-year change would give, and a constant shift would drop out with state fixed effects, but chain revisions vary by state and year, so the rerun is still owed.

In plain English. Our AI series is the same as Fang's: the means differ by 0.0006 SD, and the minimum and maximum match too. Renewable use matches only on the June 2023 data, which is how we know Fang used the old accounting.

EPU (Economic Policy Uncertainty index). Our mean ln EPU is 0.71 below Fang's Table A1 mean. But Fang's own Table 2 constants imply a mean of about 5.3, close to ours (5.38), so Table A1 may describe a different version of the index than the regressions use. A constant ratio between the two indexes would drop out with state effects, but the gap isn't constant: our minimum is 0.090 below Fang's and our maximum 0.96 below. So we can't say the difference drops out.

Pending: the GDP rerun in 2012 dollars, flagged in the table note.

5.3 The bridge from Fang's design to ours

The question. Which design difference makes Fang's result change? We move from Fang's model to ours one step at a time.

Each step adds one change to Fang's equation: (1) year fixed effects δ_t; (2) the sample that has a lagged value; (3) last year's AI instead of this year's; (4) standard errors clustered by state; (5) a longer sample; (6) dropping DC. The final step is our H1 design with Fang's outcome:

$$\ln RE_{st} = \beta \ln AI_{s,t-1} + \lambda \ln EPU_{st} + \gamma' X_{st} + \alpha_s + \delta_t + \varepsilon_{st}$$

Analogy. Changing one ingredient at a time to find out which one made the cake fall.

What "yes" and "no" would look like. If Fang's association were robust, the coefficient would stay positive at every step.

Table 18. Bridge from Fang's model to ours on today's EIA (U.S. Energy Information Administration) SEDS (State Energy Data System): AI coefficient (elasticity) at each step.

Step	AI postings	p-value	Data-center jobs	p-value
Fang's model, today's data	−0.0188	p = 0.317	−0.0655	p = 0.229
+ year effects	−0.109	p = 0.004	−0.0628	p = 0.262
+ lag sample	−0.105	p = 0.039	−0.0242	p = 0.749
+ last year's AI	−0.0761	p = 0.112	−0.112	p = 0.110
+ clustered errors	−0.0761	p = 0.148	−0.112	p = 0.193
+ longer sample	−0.0496	p = 0.094	−0.0279	p = 0.578
+ drop DC (our design)	−0.0479	p = 0.109	−0.0567	p = 0.208

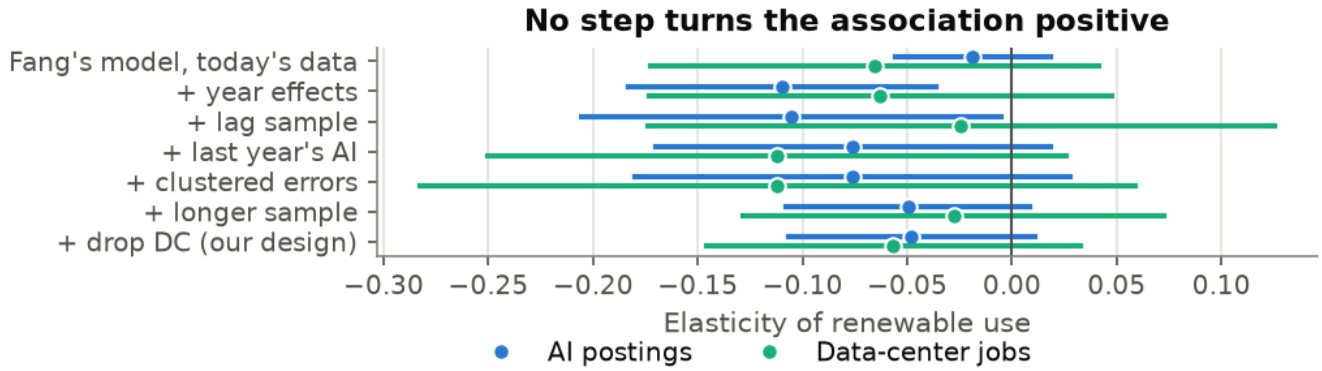


Figure 9. AI coefficient at each step of the bridge, with 95% intervals, for both AI measures.

In plain English. On today's data, Fang's model already gives a small negative number, and adding year effects makes it clearly negative (p = 0.004). By our final design neither measure is significant: postings −0.0479 (p = 0.109) and data-center jobs −0.0567 (p = 0.208). Data-center jobs are the more reliable measure for longer samples, because almost every postings year comes from a different AI Index edition.

What we can rule out. In our design, per typical within-state swing in log postings, renewable use moves between −2.6% and 0.27%; for data-center jobs, between −3.2% and 0.70%.

5.4 The trend test and placebo industries

The question. Is Fang's association specific to AI, or would any growing series show it?

$$\ln RE_{st} = \beta \ln AI_{st} + \tau t + \lambda \ln EPU_{st} + \gamma' X_{st} + \alpha_s + \varepsilon_{st}$$

Placebo: replace $\ln AI_{st}$ with $\ln \text{Jobs}_{st}^{\text{pet care}}$ (or distillery jobs).

Table 19. Symbols in the trend and placebo tests; the rest is Fang's model.

Symbol	Plain meaning
τt	A straight-line national trend in years
$\ln \text{Jobs}^{\text{pet care}}$	Log jobs in pet care (except veterinary), from BLS (Bureau of Labor Statistics) QCEW (Quarterly Census of Employment and Wages). Distilleries are a second placebo

Analogy. If your "AI detector" also beeps for pet groomers and whiskey makers, it's detecting growth, not AI.

What "yes" and "no" would look like. If Fang's result were about AI: the AI coefficient survives a trend, and the placebo industries show nothing.

Table 20. Fang's column 6 on the June 2023 SEDS (State Energy Data System) data that reproduce Fang: trend test and placebo industries.

Version (June 2023 SEDS)	Coefficient	95% interval	p-value	State-years
Fang's model, AI postings	0.0348	−0.0011 to 0.071	p = 0.057	204
+ linear trend, AI postings	−0.0419	−0.095 to 0.011	p = 0.121	204
Pet-care jobs instead of AI	0.298	0.10 to 0.50	p = 0.003	202
Distillery jobs instead of AI	0.100	0.0048 to 0.20	p = 0.040	135

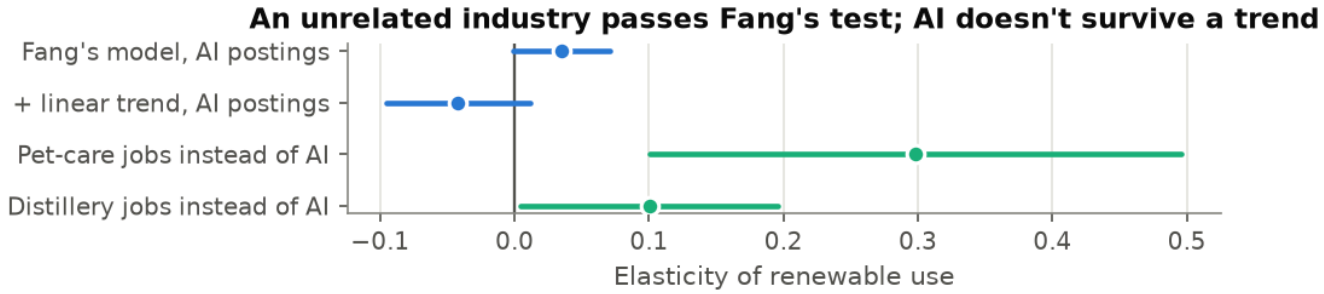


Figure 10. Trend test and placebo industries in Fang's model on June 2023 SEDS, with 95% intervals. Green rows are placebo industries.

In plain English. A simple trend turns Fang's positive coefficient negative (−0.0419, p = 0.121), and pet-care jobs are significantly associated with renewables in Fang's model (0.298, p = 0.003), while AI postings are not once a trend is included (p = 0.121). The two coefficients belong to different variables, so their sizes can't be compared. Fang's model picks up anything that grew over 2018-2021.

What we can rule out. That the association is specific to AI in Fang's design.

5.5 EPU moderation

The question. Is the AI association stronger where policy uncertainty is higher, as Fang report?

$$\ln RE_{st} = \beta \ln AI_{st} + \lambda \ln EPU_{st} + \phi \left(\ln AI_{st} \times \ln EPU_{st}\right) + \gamma' X_{st} + \alpha_s + \varepsilon_{st}$$

Table 21. Symbol added for EPU (Economic Policy Uncertainty index) moderation; the rest is Fang's model (our version adds year effects and a lag).

Symbol	Plain meaning
ϕ	How the AI association changes as EPU rises (the moderation)

Analogy. Does an umbrella sell better on rainy days? The interaction asks whether AI "sells" renewables better when policy is uncertain.

What "yes" and "no" would look like. "Yes": ϕ clearly nonzero in both designs. "No": significant only without year effects.

Table 22. AI postings x state EPU (Economic Policy Uncertainty index), today's SEDS (State Energy Data System).

Design	Interaction ϕ	95% interval	p-value
Fang's design	0.0193	0.0071 to 0.032	p = 0.002
Our design	0.0117	−0.0049 to 0.028	p = 0.163

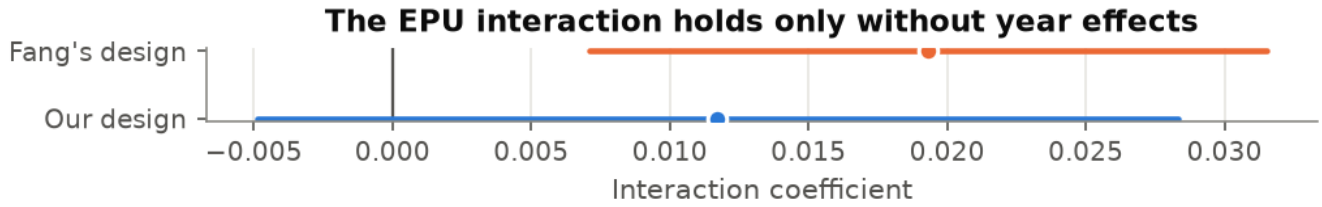


Figure 11. AI x EPU interaction in each design, with 95% intervals.

In plain English. In Fang's design the interaction is positive (p = 0.002); in ours it isn't significant (p = 0.163). Like the main result, it depends on leaving out nationwide year effects.

What we can rule out. In our design, an interaction above 0.028.

5.6 Results by renewable type, subgroups and Supplement B

The question. Do Fang's findings by type of renewable, by group of states, and for fossil fuels hold up? Each uses Fang's equation with a different outcome (wind, solar, hydro, geothermal, biomass; fossil fuels) or on a subset of states (high vs low price; net producers vs consumers):

$$\ln RE_{st}^{\text{type}} = \beta \ln AI_{st} + \lambda \ln EPU_{st} + \gamma' X_{st} + \alpha_s + \varepsilon_{st}$$

Analogy. If a medicine works, it should work in each age group, not only in the pooled study. **"Yes"** would be consistent results across types and groups; **"no"** is results that come and go with the specification.

Table 23. AI postings and each renewable type, today's SEDS (State Energy Data System) (elasticities).

Renewable type	Fang's design	p-value	Our design	p-value
Wind	0.0836	p = 0.078	−0.0107	p = 0.860
Solar	0.383	p < 0.001	0.119	p = 0.411
Hydro	0.0212	p = 0.615	0.0142	p = 0.821
Geothermal	0.0414	p = 0.543	0.197	p = 0.285
Biomass	−0.113	p < 0.001	−0.0785	p = 0.016

By type, solar is strongly positive in Fang's design (p < 0.001) and not in ours (p = 0.411): solar grew everywhere over these years, and year effects absorb that.

Subgroups. Fang split states into groups (Tables 3-5) and summarize the results in their Table A3. Who counts as a "net producer" depends on the definition: our rule gives 120 producer state-years against Fang's 76 (their Table 3). The table below checks each Table A3 claim on the June 2023 data, the version that reproduces Fang's main result, with AI postings and Fang's mean splits.

Table 24. Fang's Table A3 subgroup claims against our estimates on SEDS (State Energy Data System) as published in June 2023 (AI postings, Fang's model).

Fang's Table A3 claim: AI is associated with renewables in	Fang's estimate for that group	Ours, June 2023 SEDS	Reproduces?
High electricity prices	+0.0628 (SE 0.03, N 108)	+0.0516 (p=0.133)	Doesn't (same ordering, not significant)
Low electricity consumption	+0.0730 (SE 0.028, N 100)	+0.0602** (p=0.024)	Reproduces
Net electricity importers	+0.0425 (SE 0.024, N 128)	-0.0513 (p=0.112)	Doesn't (does not match)
High income	+0.1236 (SE 0.034, N 96)	+0.1211*** (p=0.001)	Reproduces
Low tech (low R&D (research and development) funding)	+0.0704 (SE 0.028, N 104)	+0.0367* (p=0.075)	Reproduces
Frequent extreme weather	+0.0790 (SE 0.03, N 104)	-0.0060 (p=0.782)	Doesn't (does not match)
West	+0.1176 (SE 0.05, N 52)	+0.0723 (p=0.119)	Doesn't (does not match)
Midwest	+0.0626 (SE 0.03, N 48)	+0.0717*** (p=0.009)	Reproduces

Stars: * p < 0.1, **p < 0.05,** * p < 0.01. "Reproduces" means the same ordering and significance as Fang's groups. Exploratory: the June 2023 rows were added after seeing Table 2. Full results, including median splits and data-center jobs: results/2026-11-07/fang/tables/fang8_splits_table.csv.

Supplement B. Fang's supplement reports robustness checks in Tables B1 (other measures, including fossil fuels, which should *not* rise with AI) and B2 (winsorized data and extra controls). We ran each on both data versions.

Table 25. Fang's Supplement Tables B1 and B2 next to ours: AI coefficient (p-value), Fang's model, 2018-2021.

Fang's check	Fang's AI coefficient	Ours, June 2023 SEDS (State Energy Data System)	Ours, today's SEDS
B1 (1): AI share of job postings	0.0605 (t 2.38)	0.0411 (p = 0.076)	−0.0242 (p = 0.309)

Fang's check	Fang's AI coefficient	Ours, June 2023 SEDS (State Energy Data System)	Ours, today's SEDS
B1 (2): renewable production instead of use	0.0605 (t 2.38)	0.0461 (p = 0.031)	−0.00182 (p = 0.939)
B1 (3): coal use	−0.178 (t −1.43)	−0.119 (p = 0.208)	−0.119 (p = 0.208)
B1 (4): natural gas use	0.00230 (t 0.13)	0.0134 (p = 0.298)	0.0160 (p = 0.228)
B1 (5): petroleum use	−0.0160 (t −1.08)	0.00620 (p = 0.511)	0.00672 (p = 0.479)
B2 (1): winsorized	0.0374 (t 1.99)	0.0346 (p = 0.059)	−0.0178 (p = 0.339)
B2 (2): + energy intensity	0.0444 (t 2.23)	0.0318 (p = 0.085)	−0.0201 (p = 0.288)
B2 (3): + HDD (heating degree days)	0.0438 (t 2.18)	0.0145 (p = 0.495)	−0.0131 (p = 0.540)
B2 (4): + CDD (cooling degree days)	0.0427 (t 2.12)	0.0113 (p = 0.612)	−0.0210 (p = 0.345)
B2 (5): + energy spending per person	0.0442 (t 2.19)	0.0112 (p = 0.616)	not run

Fang's B1 columns 1 and 2 report identical numbers in every cell (coefficients, t statistics and constants), so one is likely a duplicated column. B2 columns 3-5: our degree days cover the 48 contiguous states (192 state-years against Fang's 203). B2 column 5 on today's SEDS isn't run: today's expenditure data aren't in our files. Pre-registered in EXPERIMENTS.md section 14.7.

In plain English. On the June 2023 data, our estimates are positive and Fang's values fall inside our 95% intervals in every column of B1 and B2, though the coefficient shrinks and loses significance once degree days are added for the 48 contiguous states. On today's SEDS, every comparable estimate is negative or near zero, and Fang's values fall outside our intervals. The fossil-fuel checks show no significant association on either version, in Fang's results or ours.

Verdict: why Fang found a positive association and we don't. Three things. **The data version:** EIA's 2023 change in how SEDS counts wind, solar, hydro and geothermal shrank renewable energy in a way that differs by state; on the old data we get most of Fang's number back. **Year effects:** renewables and AI postings both rose nationally in 2018-2021, so without year effects any growing series, including pet-care jobs, looks associated with renewables. **Timing and sample:** with last year's AI, clustered errors and a longer sample, neither AI measure is significant. Fang's data were accurate for the version available then (Table A1 matches ours on that version); what differs is the data version and the design.

6. Extensions

All extensions use the H1 design with a different outcome or AI measure, so the equation is H1's: $y_{st} = \beta AI_{s,t-1} + \text{controls} + \alpha_s + \delta_t + \varepsilon_{st}$. A result counts as a finding only if its Benjamini-Hochberg q-value is below 0.10 within its family of tests and next year's AI doesn't explain it as well. **Analogy:** running many tests is like buying many lottery tickets, and the q-value adjusts for how many tickets we bought. **"Yes"** would be a q-value below 0.10 with a clean timing test; **"no"** is everything else.

6.1 AI by renewable type

Table 26. Selected by-type rows; the full family is in results/2026-11-07/extend/s2_bytype_results.csv.

Outcome and AI measure	Coefficient	p-value	q-value
Wind generation (log), AI postings	−0.0477	p = 0.002	0.086
Wind generation (log), data-center jobs	−0.181	p = 0.016	0.147
Solar share, data-center jobs	−0.0000593	p = 0.984	0.984
Solar generation (log), data-center jobs	0.0420	p = 0.746	0.777
Solar additions per 1,000 MW, data-center jobs	−0.599	p = 0.755	0.777

In plain English. Only 1 by-type row passes the multiple-testing bar: AI postings and wind generation (−0.0477, q = 0.086). It runs against the expected sign and loses significance with state trends, so it's exploratory. Solar shows nothing.

6.2 What AI is associated with

Table 27. What data-center jobs are associated with: selected outcomes from the 30-row family.

Outcome (data-center jobs)	Coefficient	p-value	q-value
Commercial electricity sales per person (log)	−0.00372	p = 0.764	0.929
Industrial sales per person (log)	−0.00493	p = 0.714	0.929
Residential sales per person (log)	0.000576	p = 0.824	0.929
Commercial price (cents per kWh)	0.220	p = 0.342	0.803
Gas share of generation	0.00343	p = 0.588	0.929
Wind added per 1,000 MW	1.89	p = 0.015	0.436
Gas added per 1,000 MW	−2.75	p = 0.045	0.673
Net electricity imports (share of sales)	0.000773	p = 0.966	0.966
Power-sector CO2 (carbon dioxide) (log)	−0.00516	p = 0.836	0.929

In plain English. None of the 30 rows (15 outcomes, two AI measures) passes the bar; the smallest q-value is 0.436. Per typical within-state swing in data-center jobs, commercial electricity sales per person move between −1.6% and 1.2%.

6.3 What drove the national renewables trend

Year effects absorb everything national. To ask what's inside them, we replace δ_t with measured national drivers: RPS (Renewable Portfolio Standard) targets, solar and wind costs, the ITC (Investment Tax Credit) and PTC (Production Tax Credit) from CRS (Congressional Research Service) reports and federal law, and natural gas prices:

$$y_{st} = \beta AI_{s,t-1} + \sum_k \kappa_k D_{kt} + \text{controls} + \alpha_s + \varepsilon_{st}$$

where D_{kt} are the drivers and κ_k their associations. **Analogy:** opening the "national weather" box to see whether it's sun, wind or rain. **"Yes" for AI** would be a coefficient that changes a lot once the drivers replace the year effects.

Table 28. The AI coefficient barely moves across ways of handling the national trend.

Model (renewable share)	AI coefficient	p-value
State effects only	−0.0124	p = 0.046
Measured drivers	−0.0111	p = 0.073
State and year effects (H1)	−0.0101	p = 0.100

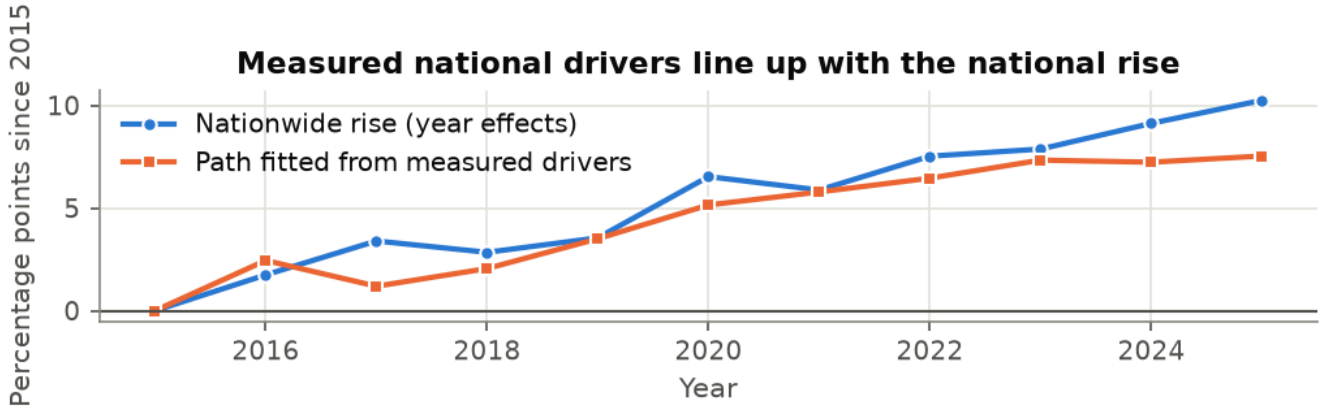


Figure 12. The national rise in the renewable share (year effects, relative to 2015) and the path fitted from measured national drivers.

In plain English. The AI coefficient is nearly the same whether we use state effects only (−0.0124), measured drivers (−0.0111) or year effects (−0.0101), so the national rise isn't loading on AI. The drivers' fitted path correlates with the year effects at 0.95. That's consistent with costs, tax credits and mandates driving the trend, but not proof: any rising series correlates with a rising trend.

6.4 The AI index and the relevance test

A broader AI measure. Jobs are one window on AI. We also built an "infrastructure + talent" index from QCEW (Quarterly Census of Employment and Wages) data-center jobs, establishments and wages, AI postings, and OEWS (Occupational Employment and Wage Statistics) research scientists (2018-2023, 43 states). Its parts hang together (Cronbach's alpha 0.73), and it correlates at 0.55 with the BTOS (Business Trends and Outlook Survey) 2024 share of firms using AI. Across its 136 rows per sample, none passes the bar (smallest q-value 0.863).

The relevance test. Before asking what AI does to renewables, we check whether our AI measure tracks electricity demand at all: $\ln \mathrm{CommercialSales}_{st} = \beta AI_{s,t-1} + \mathrm{controls} + \alpha_s + \delta_t + \varepsilon_{st}$, pooled and with a separate slope for each state. **Analogy:** before using a thermometer to study fevers, check that it moves when you hold it in warm water. **"Yes"** would be a clearly positive pooled slope.

Table 29. Commercial electricity sales per person on data-center jobs: pooled slope and per-state slopes.

Version	Slope	Rank among 50 states	Note
Pooled, all states	−0.0037		p = 0.764
Virginia	0.24	3	Descriptive: one state
Texas	0.098	7	Descriptive: one state
Georgia	0.020	21	Descriptive: one state

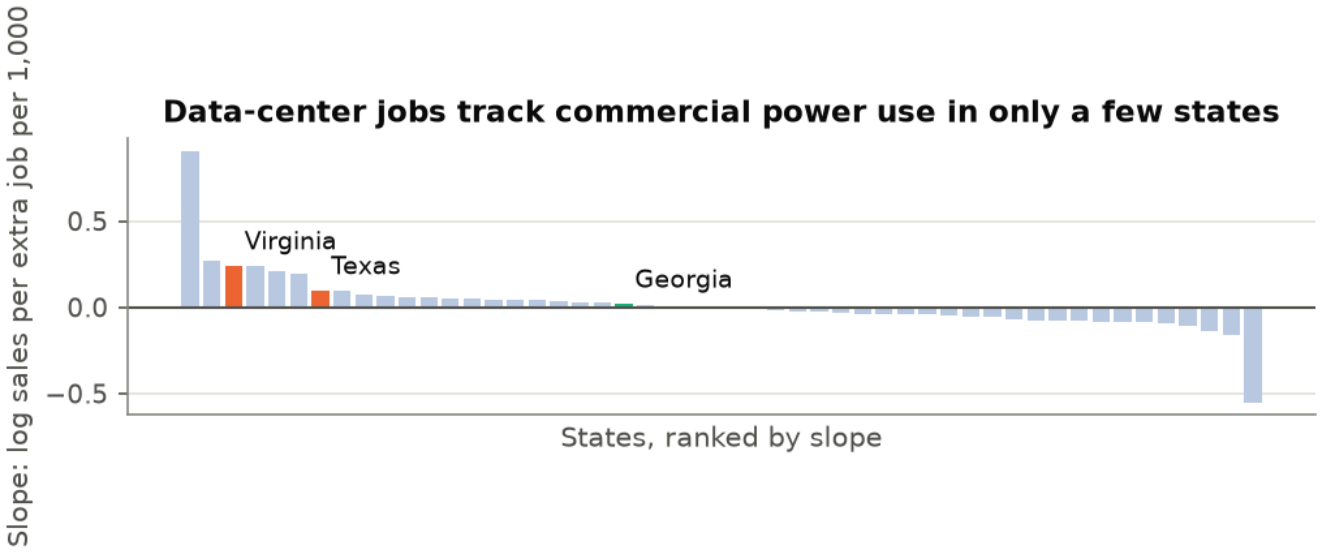


Figure 13. Each state's own slope of commercial electricity sales per person on data-center jobs, ranked. Virginia and Texas stand out; Georgia sits in the middle.

In plain English. Pooled across states, data-center jobs don't track commercial electricity use (p = 0.764). They do in Virginia (rank 3 of 50) and Texas (rank 7), the two largest data-center markets. Georgia's own slope is small (rank 21). So for most states our AI measure is a weak window on data-center power demand, which limits what any of our models can see.

Groupings. We also grouped states by AI investment, data-center jobs per TWh (terawatt-hours) of sales and land per TWh. No grouping difference survives the multiple-testing correction on all 50 states (smallest q-value 0.158); the land-based grouping that looked significant is driven by Alaska.

7. Does data-center demand drive more solar? A team hypothesis

The hypothesis is a chain with three links: **data-center growth → more electricity demand → utilities add capacity → more of that capacity is solar.** If all three links hold, states with faster data-center growth should add more solar. We check each step with the H1 design ($y_{st} = \beta AI_{s,t-1} + \text{controls} + \alpha_s + \delta_t + \varepsilon_{st}$). **Analogy:** a row of dominoes falls only if every domino hits the next. **"Yes"** would be a clearly positive association at every link, with clean timing.

Table 30. Each step of the chain checked against data-center jobs (the AI measure), 50 states, 2015-2025.

Link	What we measured	Coefficient	p-value	Reading
1. AI jobs → demand	Commercial electricity sales per person (log)	−0.00372	p = 0.764	Not in pooled data; only Virginia and Texas
1. AI jobs → demand	Total electricity use per person (log, H1)	−0.00157	p = 0.795	No association
2. AI jobs → new capacity	New renewable capacity, MW (H1)	70.8	p = 0.033	Positive in MW, but not per 1,000 MW, and the lead is positive too
2. AI jobs → new capacity	Gas added per 1,000 MW	−2.75	p = 0.045	Fails multiple testing (q = 0.673)
3. AI jobs → solar	Solar added per 1,000 MW	−0.599	p = 0.755	No association
3. AI jobs → solar	Solar share of generation	−0.0000593	p = 0.984	No association

We can't estimate the demand → capacity link directly; we can only check whether faster data-center growth goes with each step.

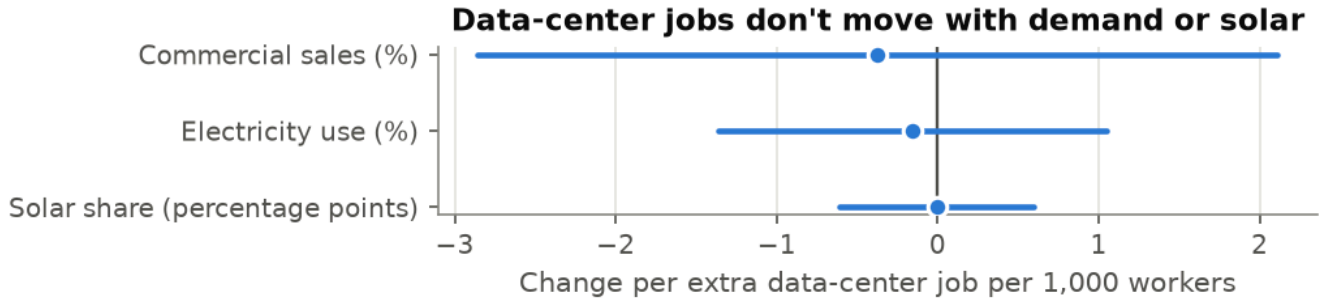


Figure 14. Associations at the demand link and the solar link, with 95% intervals.

In plain English. The first step doesn't show up in our data: across states, data-center jobs aren't associated with more electricity use, except in Virginia and Texas. And at the end of the chain, solar shows no association at all (p = 0.984).

Why the hypothesis could still be true but invisible here.

- **Timing.** Most data-center power demand arrives after 2022, and much of it after our data end in 2025. Georgia Power's large-load commitments alone went from 3,600 MW in October 2023 to 15,600 MW across 32 customers on June 30, 2026 (Q2 2026 large-load report, p. 4 [PDF p. 8]).
- **Purchases across state lines.** Tech companies often buy solar power from projects in other states. Our outcomes count solar where it's built, not where it's bought.
- **Planned, not built.** Much of the response is still in plans and interconnection queues, which our capacity data (built plants only) don't show.
- **National effects.** If data-center demand lowers solar costs or raises solar build-out everywhere, year effects absorb it, and state comparisons can't see it.

What would test it better: interconnection queue data by state (projects waiting to connect, such as LBNL (Lawrence Berkeley National Laboratory)'s *Queued Up*); utility resource plans and their stated reasons; an event study around 2022 comparing states before and after large data-center announcements; and data-center megawatts by state instead of jobs.

8. Georgia

Georgia is a useful case: a fast-growing data-center market served mostly by one regulated utility, Georgia Power, whose plans are public through the Georgia PSC (Public Service Commission).

AI growth. Georgia ranked 22 of 50 states in data-center jobs per 1,000 workers in 2015 and 2 in 2025; its rise over 2015-2024 ranks 2. On AI job postings it ranks only 25 in 2025, a reminder that the two AI measures capture different things.

The solar boom. Solar's share of Georgia's in-state generation rose from 0.1% in 2015 to 7.7% in 2025.

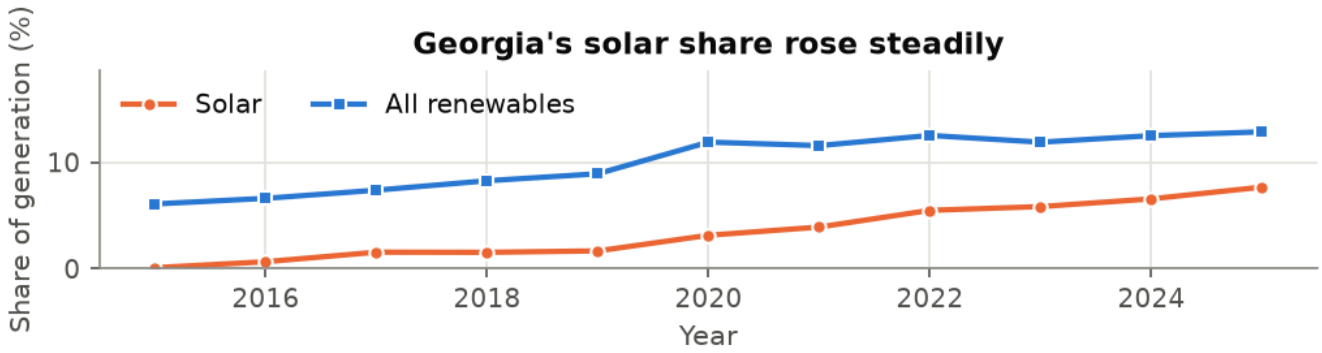


Figure 15. Solar and total renewable share of Georgia's in-state utility-scale generation.

What our estimates imply for Georgia. Taking our national solar-share association and Georgia's actual rise in data-center jobs, the association could account for at most 25% of Georgia's solar-share rise, using the end of the confidence interval most favorable to AI, and the point estimate has the opposite sign. Georgia's solar boom coincides with falling costs, tax credits and the utility's procurement plans; we didn't estimate those for Georgia, and our data don't link the boom to AI.

Table 31. Georgia Power's figures, each with what it measures and one cite (printed page, PDF page in brackets). Documents: 2025 IRP (Integrated Resource Plan) = Georgia Power 2025 IRP, main document (Georgia PSC (Public Service Commission) Docket 56002); IRP summary = the plan's one-page summary; Q2 2026 report = Q2 2026 Large Load Economic Development Report (Docket 55378).

Item	Figure	What it measures	Document	Page
Load growth, IRP forecast	about 8,200 MW	Risk-adjusted load growth forecast from winter 2024/2025 through winter 2030/2031	2025 IRP	p. 1 [PDF p. 8]
Load growth, approval summary	about 8,500 MW	Load growth over the next six years, as stated in the summary issued after the plan's approval	IRP summary	p. 1 [PDF p. 1]
Commitments in October 2023	3,600 MW	Commitments at the October 2023 IRP Update filing, the baseline both 2026 reports compare against	Q2 2026 report	p. 4 [PDF p. 8]
Large-load commitments	15,600 MW across 32 customers	Customer commitments as of June 30, 2026, for load scheduled through the mid-2030s	Q2 2026 report	p. 1 [PDF p. 5]
Commitments due by winter 2028/2029	8,700 MW	The near-term part of those commitments: load scheduled by winter 2028/2029	Q2 2026 report	p. 1 [PDF p. 5]
Large-load pipeline	84,800 MW	Economic-development pipeline through the mid-2030s, including prospects without commitments	Q2 2026 report	p. 1 [PDF p. 5]

Item	Figure	What it measures	Document	Page
Capacity requests for proposals	up to 9,500 MW	Active requests for proposals for capacity of any fuel, by the end of 2030	2025 IRP	p. 3 [PDF p. 10]
Renewables, active requests	more than 3,500 MW	Renewable resources in active requests for proposals, by the end of 2030	2025 IRP	p. 3 [PDF p. 10]
Renewables, new requests	up to 4,000 MW	New requests for proposals for energy from renewable resources, by 2035	2025 IRP	p. 60 [PDF p. 67]
Batteries and combustion turbines	more than 2,065 MW	Battery storage and combustion turbines combined, added by the end of 2027	2025 IRP	p. 3 [PDF p. 10]
Planned company-owned batteries	about 780 MW, including 500 MW certified in 2024	Nameplate capacity of the seven planned battery projects in Attachment C.3, in service 2026-2027 (our sum of the table rows)	2025 IRP	p. 128 [PDF p. 135]
New combustion turbines	1,323 MW (three units of 441 MW)	Plant Yates Units 8-10, company-owned gas/oil turbines, summer capacity, in service 12/2026-08/2027	2025 IRP	p. 126 [PDF p. 133]
Coal units kept running	about 1,100 MW	Extended operation of coal units at Plants Scherer and Gaston	2025 IRP	p. 2 [PDF p. 9]

These are the utility's own figures, quoted from the filings and checked against each page (results/2026-11-09/georgia/irp_figures.csv); they aren't our estimates. The two load-growth figures cover different periods. The 2,065 MW is batteries and combustion turbines together.

Georgia Power's response. The 2025 IRP (Integrated Resource Plan) forecasts about 8,200 MW of load growth from winter 2024/2025 through winter 2030/2031 (2025 IRP, p. 1 [PDF p. 8]). That forecast includes the utility's pipeline of potential and committed large-load customers (same page), and data centers are the largest segment of committed large load in the Q2 2026 report's attachment (19 projects, 9,149 MW with a Contract for Electric Service). The summary issued after the plan's approval gives about 8,500 MW over the next six years (IRP summary, p. 1 [PDF p. 1]). The plan responds with a mix: renewables (more than 3,500 MW in active requests by 2030, p. 3 [PDF p. 10]; up to 4,000 MW in new requests by 2035, p. 60 [PDF p. 67]), batteries (about 780 MW, including 500 MW certified in 2024, p. 128 [PDF p. 135]), new combustion turbines (1,323 MW (three units of 441 MW), p. 126 [PDF p. 133]) and coal units kept running (about 1,100 MW, p. 2 [PDF p. 9]). So the utility's response to data-center demand includes solar, but not only solar.

The solar cost projection to 2035

The question. If solar keeps getting cheaper as the world builds more of it, what would new Georgia solar cost in 2035?

$$C_t = C_{base} \times \left(\frac{X_t}{X_{base}} \right)^b, \qquad b = \log_2(1 - LR)$$

The base. The starting point is Georgia's 17 utility-scale solar projects that began operating in 2023-2025: C_{base} is their capacity-weighted LCOE (levelized cost of energy), 63.81 dollars per MWh, and X_{base} is global solar capacity in each project's start year, weighted the same way, 1,936.3 GW (gigawatts) (roughly mid-2024). The 2025 column below differs by learning rate because global capacity had already

grown from that base to 2,378 GW in 2025, and each learning rate turns that growth into a different cost drop.

Table 32. Symbols in the learning-curve projection.

Symbol	Plain meaning
C_t, C_{base}	LCOE (levelized cost of energy) of new utility-scale solar in Georgia in year t and at the base, 2025 dollars per MWh, without tax credits
X_t, X_{base}	Cumulative global solar PV (photovoltaic) capacity in year t (IEA (International Energy Agency) WEO (World Energy Outlook) 2025 scenario paths) and at the base
LR	Learning rate: the cost drop for each doubling of cumulative capacity
b	The curve's slope, set by the learning rate

Analogy: TVs got cheaper as factories made more of them; the same pattern has held for solar panels. **"Yes"** in this context means only that the arithmetic gives a lower cost; this is a projection under assumptions, not a test or a forecast of ours.

Table 33. Georgia solar LCOE (levelized cost of energy), 2025 dollars per MWh. Paths from IEA (International Energy Agency) WEO (World Energy Outlook) 2025, Annex A: Low = Current Policies (Table A.3a, p. 417), Central = Stated Policies (Table A.3b, p. 423), High = Net Zero Emissions by 2050 (Table A.3c, p. 429).

Learning rate	Capacity path	2025	2035	Change 2025-2035 (%)
12%	Low	61.44	50.40	−18.0
24% (main)	Low	58.82	38.45	−34.6
24% (main)	Central	58.82	37.06	−37.0
24% (main)	High	58.82	32.33	−45.0
45% (2014-2020 epoch)	High	53.45	14.51	−72.9

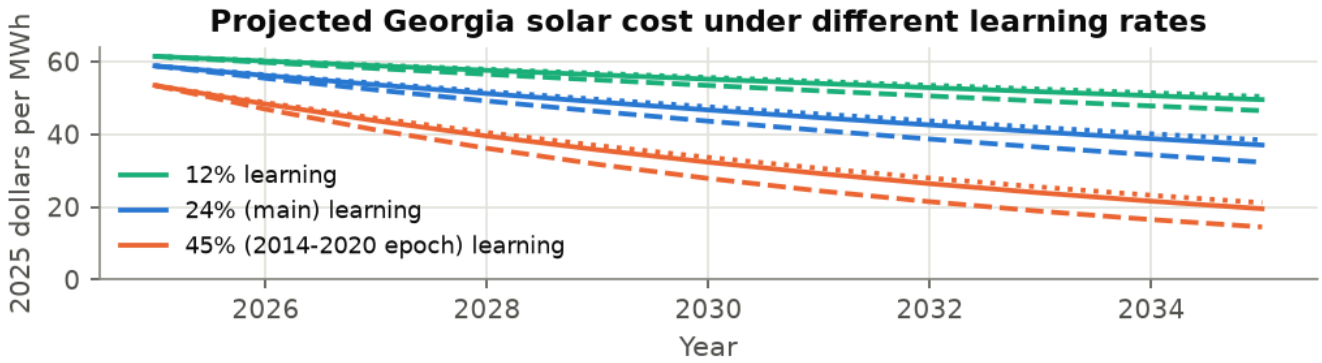


Figure 16. Learning-curve arithmetic for Georgia solar. Solid: central capacity path; dotted: low; dashed: high.

In plain English. Starting from Georgia's capacity-weighted LCOE of 63.81 dollars per MWh for 17 recent projects, a 24% learning rate on the central path implies about 37.06 dollars per MWh in 2035; across all twelve combinations, the range is 14.51 to 50.40. These aren't forecasts of ours and contain no AI variable. One warning: national solar costs actually rose from 50.53 to 61.45 dollars per MWh between 2022 and 2025, so the curve assumes costs resume falling.

9. Limitations

- **Jobs aren't megawatts.** Data-center jobs in NAICS 518210 (North American Industry Classification System code 518210, Computing Infrastructure Providers, Data Processing, Web Hosting) include general data processing and web hosting, not only AI, and a data center's power use isn't proportional to its staff. A highly automated facility can use a lot of electricity with few workers.
- **Little variation within states.** Our models compare each state with itself, and data-center jobs move only modestly within a state (one SD is 0.521 jobs per 1,000 workers). That limits how precisely any effect can be measured.
- **Timing.** Our data end in 2025, before most of the data-center build-out now being planned.
- **No strong instrument.** Our only instrument is weak, so we can't separate cause from selection.
- **Utility-scale only.** EIA (U.S. Energy Information Administration) state generation misses rooftop solar.
- **The EIA accounting change** affects every SEDS (State Energy Data System)-based renewable number, and results differ by data version.
- **Weather** enters one check only, for the 48 contiguous states.
- **Georgia** is regulated, so its utility's behavior may not generalize to states with competitive power markets.
- **Pending:** the rerun of Fang's model with GDP in chained 2012 dollars (section 5.2). BEA (Bureau of Economic Analysis)'s archive of the 2012-dollar data has to be downloaded by a person.

10. FAQ

Is a null result a failure? No. A precise null rules things out. Our intervals exclude, for example, a 1% rise in electricity use per person from a typical swing in data-center jobs. Knowing what *didn't* happen in 2015-2025 is useful when people predict what AI will do to the grid.

Why year fixed effects? Because 2015-2025 saw big national changes: tax credits, falling solar prices, COVID. Without year effects, any state series that rose over time would look associated with renewables. Section 5.4 shows exactly this: pet-care jobs "predict" renewables in a model without year effects.

Why didn't we get Fang's result? Mostly the data version and the missing year effects. On the EIA (U.S. Energy Information Administration) data Fang used, we get most of their number back; with year effects, a lag and a longer sample, it isn't positive.

Why so many data sources? No agency measures AI, electricity and policy together. Each source answers one piece, and each is the standard source for its piece.

Could AI still matter? Yes. Our results describe 2015-2025 and data-center *jobs*. The data-center build-out now under way is far larger: Georgia Power's large-load pipeline alone is 84,800 MW through the mid-2030s (Q2 2026 large-load report, p. 1 [PDF p. 5]). Megawatt-based data and a few more years could show what jobs can't.

11. Glossary

Acronyms and technical terms, A to Z.

ACS (American Community Survey). The Census Bureau's annual household survey; we use its broadband table.

AI (artificial intelligence). Computer systems that perform tasks such as prediction and language; here measured through data-center jobs or AI job postings.

Association. Two things moving together. Not proof that one causes the other.

BEA (Bureau of Economic Analysis). The federal agency that publishes state GDP, income and population estimates.

Benjamini-Hochberg q-value. A p-value adjusted for running many tests at once; we require q below 0.10.

BLS (Bureau of Labor Statistics). The federal agency that counts jobs and wages.

BTOS (Business Trends and Outlook Survey). A Census Bureau survey that asks firms, among other things, whether they use AI.

Btu (British thermal units). A unit of energy; EIA (U.S. Energy Information Administration) SEDS (State Energy Data System) reports energy in billions of them.

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CDD (cooling degree days). A measure of how much cooling a year needed: degrees above 65°F, summed over days.

CES (Clean Energy Standard). A state mandate like an RPS (Renewable Portfolio Standard) that also counts other low-carbon sources such as nuclear.

Clustered standard errors. Uncertainty estimates that treat a state's years as related, not independent.

CO2 (carbon dioxide). The main greenhouse gas from burning fossil fuels.

Coefficient. The slope: how much the outcome changes per unit of AI, other things fixed.

Confidence interval. The range of values the data can't rule out, at 95% confidence.

Control variable. Something held fixed in the comparison, such as income or electricity price.

CRS (Congressional Research Service). The research arm of Congress; its reports document tax-credit rules.

DC (District of Columbia). Washington, D.C., included in Fang's sample and dropped from ours.

Direct, indirect and total effects. In spatial models: a state's own association, the spillover from other states, and their sum.

EIA (U.S. Energy Information Administration). The federal statistical agency for energy; source of our electricity and SEDS data.

Elasticity. Percent change in one thing per 1% change in another.

EPU (Economic Policy Uncertainty index). A news-based index of economic policy uncertainty, by state, from Baker, Davis and Levy.

FAQ (frequently asked questions). Common questions with short answers.

Fixed effects. Terms that remove each state's permanent level (state effects) or each year's nationwide level (year effects).

GDP (gross domestic product). The value of everything a state's economy produces in a year.

GW (gigawatts). A thousand megawatts.

HDD (heating degree days). A measure of how much heating a year needed: degrees below 65°F, summed over days.

IEA (International Energy Agency). An intergovernmental energy agency; its World Energy Outlook supplies our solar capacity paths.

IRENA (International Renewable Energy Agency). An intergovernmental agency that publishes renewable capacity statistics.

IRP (Integrated Resource Plan). A utility's long-term plan for meeting demand, filed with its regulator.

ITC (Investment Tax Credit). A federal tax credit worth a share of the cost of a solar or other clean-energy project.

IV (instrumental variable). A method that uses an outside source of variation to get closer to cause and effect.

kWh (kilowatt-hour). A unit of electricity; prices are quoted in cents per kilowatt-hour.

LBNL (Lawrence Berkeley National Laboratory). A national laboratory that publishes data on renewable costs and state renewable mandates.

LCOE (levelized cost of energy). The average cost per MWh of building and running a power plant over its life.

Lead (placebo) test. Using next year's AI to check whether the timing makes sense.

Learning rate. The cost drop for each doubling of cumulative production.

Lind-Mehlum test. A test for a real curve: the slope must be clearly positive at one end and clearly negative at the other.

Mediation. Testing whether an association runs through a third variable (a channel).

Moran's I. A measure of whether neighboring states have similar values.

MW (megawatts). A unit of power plant capacity: a million watts.

MWh (megawatt-hours). A unit of electricity: one megawatt for one hour.

NAICS 518210 (North American Industry Classification System code 518210, Computing Infrastructure Providers, Data Processing, Web Hosting). The industry code for data centers, data processing and web hosting; our AI measure counts its private jobs.

NCSES (National Center for Science and Engineering Statistics). The part of NSF (National Science Foundation) that publishes science and engineering statistics.

NOAA (National Oceanic and Atmospheric Administration). The federal agency for weather and climate data; source of degree days.

NSF (National Science Foundation). The federal agency that funds research and publishes R&D (research and development) statistics.

OEWS (Occupational Employment and Wage Statistics). A BLS survey of jobs and wages by occupation; we use research scientists.

OLS (ordinary least squares). The standard regression method that fits a line by minimizing squared errors.

OWID (Our World in Data). A website that republishes research data, including global solar capacity.

p-value. The chance of a result at least this far from zero if the true value were zero.

Panel data. The same units (states) observed over many years.

PDF (Portable Document Format). A document file format; we cite filings by their PDF page.

Percentage point. The unit for differences between percentages: 10% to 12% is 2 percentage points.

Placebo industry. An industry unrelated to energy, used to check whether a model detects trends instead of AI.

PSC (Public Service Commission). The state regulator that approves a utility's plans and rates.

PTC (Production Tax Credit). A federal tax credit paid per MWh of electricity from wind and some other sources.

PV (photovoltaic). Solar panels that turn sunlight directly into electricity.

QCEW (Quarterly Census of Employment and Wages). BLS's near-complete count of jobs and wages by industry and state.

R&D (research and development). Spending on research and development.

RFP (request for proposals). A utility's call for bids to build or supply new power.

RPS (Renewable Portfolio Standard). A state law requiring a minimum share of electricity from renewables.

SD (standard deviation). A measure of typical spread around the average.

SDM (spatial Durbin model). A spatial model with both neighbors' outcomes and neighbors' AI.

SEDS (State Energy Data System). EIA's state-level estimates of energy production, use and prices for every fuel.

Shift-share instrument. A predicted value built from a state's starting share times national growth.

SLX (spatial lag of X model). A spatial model with neighbors' AI on the right-hand side.

TWh (terawatt-hours). A billion kilowatt-hours.

USPTO (U.S. Patent and Trademark Office). The federal agency that grants patents; its AI patent data were downloaded and not used.

WEO (World Energy Outlook). The IEA's annual report with scenario paths for energy to 2050.

Winsorizing. Capping extreme values at the 1st and 99th percentiles so a few outliers can't dominate.

Number check: 678 numbers traced to result files; 678 match their source and appear in the text; 0 untraced decimal numbers. Details: results/2026-11-09/explained/number_check.csv.